

Problem 1

The ABB arm has the following DH table.

Link	a	d	α	θ
1	0	d_1	$\pi/2$	*
2	a_2	0	0	*
3	a_3	0	$-\pi/2$	*
4	0	d_4	$\pi/2$	*
5	0	0	$-\pi/2$	*
6	0	d_6	0	*

- Draw the zero-angle configuration.
- Solve the inverse kinematics (show all your work).
- How many possible solutions exist?
- Write a MATLAB code to demonstrate the efficacy of your script over 100 random test cases.

Part (a)

Evaluating the DH table with all $\theta_i = 0$ (see [Problem1Figure.m](#), [Appendix A](#)):

Zero-Angle Configuration of the ABB Arm

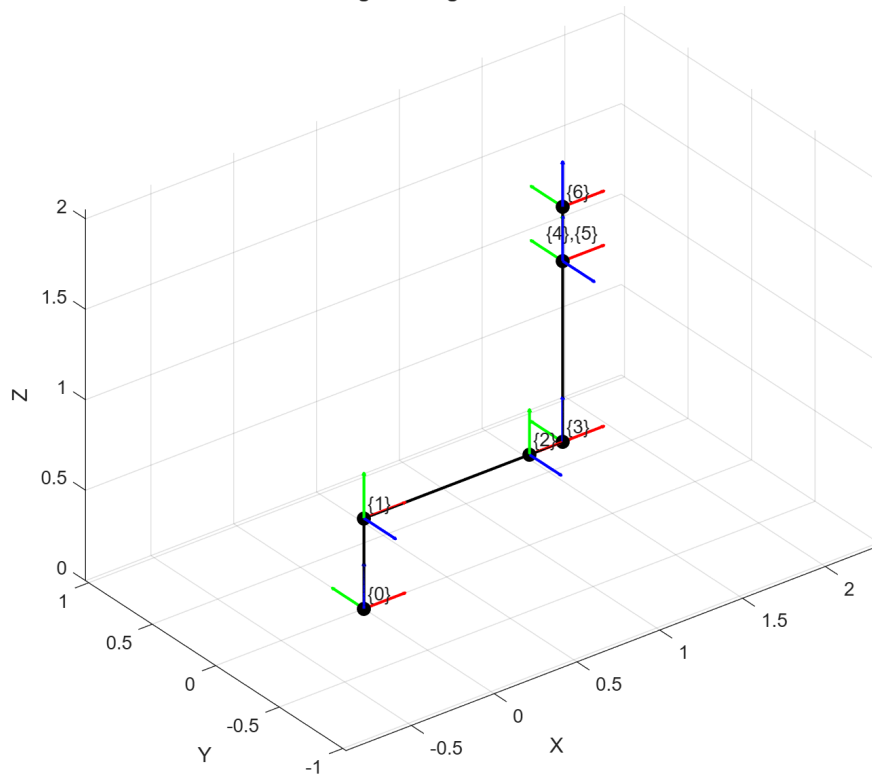


Figure 1: Zero-angle configuration of the ABB arm. Axes: x red, y green, z blue. Dimensions are illustrative and differ from Part (d) ($d_1 = 0.5$, $a_2 = 1$, $a_3 = 0.2$, $d_4 = 1$, $d_6 = 0.3$). Frames 4 and 5 share an origin, so they carry one combined label.

Part (b)

The last three axes intersect at the wrist center (origin of frames $\{4\}$ and $\{5\}$), so the position and orientation problems decouple. Let $\mathbf{P}_{06} = [P_{6x}, P_{6y}, P_{6z}]^T$ and let $\mathbf{R} = {}^0\mathbf{R}_6$ be the desired orientation. The wrist center is

$$\mathbf{P}_{05} = \begin{bmatrix} P_{5x} \\ P_{5y} \\ P_{5z} \end{bmatrix} = \mathbf{P}_{06} - d_6 \mathbf{R}(:, 3). \quad (1)$$

Joint 1. The projection of $\mathbf{P}_{05}$ onto the x_0 - y_0 plane lies at angle θ_1 from x_0 , so

$$\theta_1 = \text{atan2}(P_{5y}, P_{5x}) \quad \text{or} \quad \theta_1 + \pi. \quad (2)$$

Joints 2 and 3. With θ_1 known, the arm reduces to a planar two-link problem in the plane of $\mathbf{P}_{05}$. Define

$$r = \sqrt{P_{5x}^2 + P_{5y}^2}, \quad (3)$$

$$D = \sqrt{P_{5x}^2 + P_{5y}^2 + (P_{5z} - d_1)^2}, \quad (4)$$

$$L = \sqrt{a_3^2 + d_4^2}, \quad (5)$$

where D is the distance from O_1 to the wrist center and L is the distance from O_2 to the wrist center. The pose is reachable only if $|a_2 - L| \leq D \leq a_2 + L$. Let ψ be the elevation angle of D above the horizontal and $\beta \in [0, \pi]$ the interior angle at O_1 between a_2 and D . For θ_1 the horizontal reach is r ; for $\theta_1 + \pi$ the arm points the other way, so r is replaced by $-r$:

$$\psi = \text{atan2}(P_{5z} - d_1, \pm r) \quad (+r \text{ for } \theta_1, -r \text{ for } \theta_1 + \pi), \quad \theta_2 = \psi \pm \beta. \quad (6)$$

Law of cosines on the triangle with sides a_2 , D , L :

$$L^2 = a_2^2 + D^2 - 2a_2D \cos \beta, \quad (7)$$

$$\cos \beta = \frac{a_2^2 + D^2 - L^2}{2a_2D}, \quad \sin \beta = +\sqrt{1 - \cos^2 \beta}, \quad (8)$$

$$\beta = \text{atan2}\left(\sqrt{1 - \left(\frac{a_2^2 + D^2 - L^2}{2a_2D}\right)^2}, \frac{a_2^2 + D^2 - L^2}{2a_2D}\right). \quad (9)$$

The only sign choice is the $\pm$ in $\theta_2 = \psi \pm \beta$:

$$\theta_2 = \text{atan2}(P_{5z} - d_1, \pm r) \pm \text{atan2}\left(\sqrt{1 - \left(\frac{a_2^2 + D^2 - L^2}{2a_2D}\right)^2}, \frac{a_2^2 + D^2 - L^2}{2a_2D}\right). \quad (10)$$

For θ_3 , let δ be the fixed angle at O_2 between a_3 (the segment $O_2 \rightarrow O_3$) and L (the segment $O_2 \rightarrow$ wrist center), and γ the interior angle at O_2 between a_2 and L :

$$\delta = \text{atan2}(d_4, a_3). \quad (11)$$

Law of cosines on the same triangle, now for the angle opposite D :

$$D^2 = a_2^2 + L^2 - 2a_2L \cos \gamma, \quad (12)$$

$$\cos \gamma = \frac{a_2^2 + L^2 - D^2}{2a_2L}, \quad \sin \gamma = +\sqrt{1 - \cos^2 \gamma}, \quad (13)$$

$$\gamma = \text{atan2}\left(\sqrt{1 - \cos^2 \gamma}, \cos \gamma\right). \quad (14)$$

The elbow sign of θ_3 is tied to the sign of β in θ_2 :

$$\theta_3 = \mp(\pi - \gamma) - \delta, \quad (15)$$

with the upper sign paired with $\theta_2 = \psi + \beta$ (elbow up) and the lower sign with $\theta_2 = \psi - \beta$ (elbow down). Explicitly,

$$\theta_3 = \mp \left[\pi - \text{atan2} \left(\sqrt{1 - \left(\frac{a_2^2 + L^2 - D^2}{2a_2L} \right)^2}, \frac{a_2^2 + L^2 - D^2}{2a_2L} \right) \right] - \text{atan2}(d_4, a_3). \quad (16)$$

Joints 4, 5, and 6. From the DH table, ${}^0\mathbf{R}_3$ depends only on $\theta_1, \theta_2, \theta_3$:

$${}^0\mathbf{R}_3 = \text{Rot}_z(\theta_1) \text{Rot}_x\left(\frac{\pi}{2}\right) \text{Rot}_z(\theta_2 + \theta_3) \text{Rot}_x\left(-\frac{\pi}{2}\right) = \text{Rot}_z(\theta_1) \text{Rot}_y(-(\theta_2 + \theta_3)). \quad (17)$$

Then

$${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top {}^0\mathbf{R}_6. \quad (18)$$

From the DH table, the wrist rotation is ${}^3\mathbf{R}_6 = \text{Rot}_z(\theta_4) \text{Rot}_y(-\theta_5) \text{Rot}_z(\theta_6)$. This is a Z - Y - Z rotation with θ_5 measured about $-y$, so the signs differ from the textbook form. Multiplying out, with r_{ij} the entries of ${}^3\mathbf{R}_6$,

$$r_{13} = -\cos \theta_4 \sin \theta_5, \quad r_{23} = -\sin \theta_4 \sin \theta_5, \quad r_{33} = \cos \theta_5, \quad (19)$$

$$r_{31} = \sin \theta_5 \cos \theta_6, \quad r_{32} = -\sin \theta_5 \sin \theta_6. \quad (20)$$

Solving for the angles:

$$\theta_5 = \text{atan2} \left(\pm \sqrt{r_{13}^2 + r_{23}^2}, r_{33} \right), \quad (21)$$

$$\theta_4 = \text{atan2} \left(\frac{-r_{23}}{\sin \theta_5}, \frac{-r_{13}}{\sin \theta_5} \right), \quad (22)$$

$$\theta_6 = \text{atan2} \left(\frac{-r_{32}}{\sin \theta_5}, \frac{r_{31}}{\sin \theta_5} \right). \quad (23)$$

Singularities.

- *Shoulder* ($P_{5x} = P_{5y} = 0$): the wrist center lies on the z_0 axis and θ_1 is free.
- *Elbow* ($D = a_2 + L$ or $D = |a_2 - L|$): $\beta = 0$ (and $\gamma = 0$ or π), so the two elbow branches merge.
- *Wrist* ($\sin \theta_5 = 0$): axes 4 and 6 align and only $\theta_4 \pm \theta_6$ is determined. Choose $\theta_4 = 0$, and take θ_6 from the upper-left 2×2 block of ${}^3\mathbf{R}_6$: $\theta_6 = \text{atan2}(r_{21}, r_{11})$ for $\theta_5 = 0$ and $\theta_6 = \text{atan2}(r_{21}, -r_{11})$ for $\theta_5 = \pi$.

Part (c)

Up to 8 solutions for a reachable pose.

The count is $2 \times 2 \times 2$, one factor for each of the three sign choices:

- Shoulder: $\theta_1 = \text{atan2}(P_{5y}, P_{5x})$ or $+\pi$ (front or back), giving 2 options.

- Elbow: the $\pm\beta$ sign in θ_2 , with the matching $\mp(\pi - \gamma)$ in θ_3 (elbow up or down), giving 2 options for each shoulder choice.
- Wrist: the $\pm$ on $\sin \theta_5$ (wrist flip), giving 2 options for each arm configuration.

In degenerate cases the count changes. At the elbow singularity the two elbow branches coincide, and at the shoulder singularity θ_1 is free (infinitely many solutions). At the wrist singularity θ_4 and θ_6 are not separately determined, so each arm configuration has infinitely many wrist solutions. Outside the reachable range $|a_2 - L| \leq D \leq a_2 + L$ there are none.

Part (d)

See **Problem1.m** and **ikABB.m**, Appendix A. The test uses $d_1 = 0.290$, $a_2 = 0.270$, $a_3 = 0.070$, $d_4 = 0.302$, $d_6 = 0.072$ (m), seed 544, and a tolerance of 10^{-8} . Each of 100 random joint vectors has $\theta_1, \dots, \theta_4, \theta_6 \in [-\pi, \pi]$ and $\theta_5 = \pm(0.05 + 0.9u)\pi$ with $u \sim U[0, 1]$ and a random sign, which excludes a band of 0.05π around the wrist singularity and exercises both wrist branches. Forward kinematics gives the target pose T_{06} ; ikABB returns all 8 solutions, and each is passed through forward kinematics and compared to T_{06} . Position and rotation errors are reported separately, and the distinctness row shows that the 8 solutions are pairwise different. One separate singular case ($\theta_5 = 0$) checks the wrist-singularity branch.

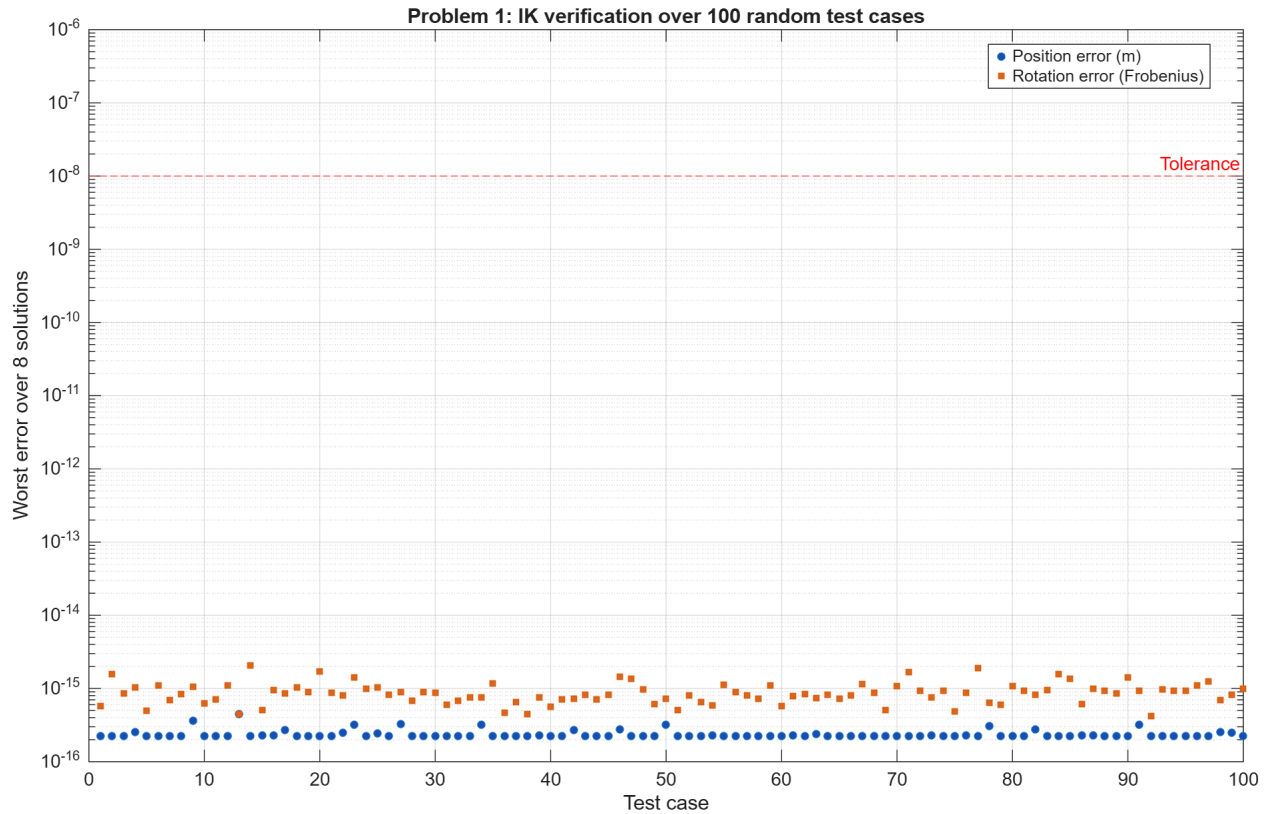


Figure 2: Worst position and rotation error over the 8 IK solutions for each of the 100 random test cases.

Appendix A: Code

Problem 1

Listing 1: Problem1.m

```

1  % Problem1: Inverse kinematics of the ABB arm, tested over random poses
2  %
3  % Arael Anaya
4  % 10920967
5  % MEGN544
6  % 10/03/2026
7
8  clear; close all; clc;
9
10 addpath(genpath('.../.../Functions'));
11
12 rng(544);
13 numTests = 100;
14 tol = 1e-8;
15 distTol = 1e-6;
16
17 d1 = 0.290; a2 = 0.270; a3 = 0.070; d4 = 0.302; d6 = 0.072;
18 dims = [d1 a2 a3 d4 d6];
19
20 DH = [
21     0      d1      pi/2      0      ;
22     a2      0      0      0      ;
23     a3      0     -pi/2      0      ;
24     0      d4      pi/2      0      ;
25     0      0     -pi/2      0      ;
26     0      d6      0      0      ];
27
28 numPoseOK = 0;
29 numRecovered = 0;
30 numDistinct = 0;
31 worstPosErr = 0;
32 worstRotErr = 0;
33 minPairDist = inf;
34 testPosErr = zeros(numTests, 1);
35 testRotErr = zeros(numTests, 1);
36
37 for test = 1:numTests
38     % theta5 is kept away from 0 and +/-pi, with a random sign so both
39     % wrist branches are exercised
40     sign5 = 2*(rand > 0.5) - 1;
41     qTrue = [ (2*rand-1)*pi ; (2*rand-1)*pi ; (2*rand-1)*pi ; ...
42              (2*rand-1)*pi ; sign5*(rand*0.9+0.05)*pi ; (2*rand-1)*pi ];
43     DH(:,4) = qTrue;
44
45     Tfk = fkChain(DH);
46     T06 = Tfk(:,:,end);
47
48     solutions = ikABB(T06, dims);

```

```

48
49 poseOK = size(solutions, 2) == 8;
50 recovered = false;
51 for k = 1:size(solutions, 2)
52     DH(:,4) = solutions(:,k);
53     Tk = fkChain(DH);
54     posErr = norm(Tk(1:3,4,end) - T06(1:3,4));
55     rotErr = norm(Tk(1:3,1:3,end) - T06(1:3,1:3), 'fro');
56     worstPosErr = max(worstPosErr, posErr);
57     worstRotErr = max(worstRotErr, rotErr);
58     testPosErr(test) = max(testPosErr(test), posErr);
59     testRotErr(test) = max(testRotErr(test), rotErr);
60     if posErr > tol || rotErr > tol
61         poseOK = false;
62     end
63
64     angErr = atan2(sin(solutions(:,k) - qTrue), cos(solutions(:,k) -
65         qTrue));
66     if norm(angErr) < 1e-6
67         recovered = true;
68     end
69 end
70 % Smallest wrapped-angle distance between any two of the solutions
71 testMinDist = inf;
72 for i = 1:size(solutions, 2)-1
73     for j = i+1:size(solutions, 2)
74         diff = solutions(:,i) - solutions(:,j);
75         testMinDist = min(testMinDist, norm(atan2(sin(diff), cos(diff)
76             )));
77     end
78 end
79 minPairDist = min(minPairDist, testMinDist);
80
81 numPoseOK = numPoseOK + poseOK;
82 numRecovered = numRecovered + recovered;
83 numDistinct = numDistinct + (size(solutions, 2) == 8 && testMinDist >
84     distTol);
85 end
86
87 % Wrist-singular test case (theta5 = 0): every returned solution must
88 % still reach the pose
89 qSing = [0.4; 0.3; -0.5; 0.7; 0; -0.2];
90 DH(:,4) = qSing;
91 Tfk = fkChain(DH);
92 T06 = Tfk(:, :, end);
93 solutions = ikABB(T06, dims);
94 singErr = 0;
95 for k = 1:size(solutions, 2)
96     DH(:,4) = solutions(:,k);
97     Tk = fkChain(DH);
98     singErr = max(singErr, norm(Tk(:, :, end) - T06, 'fro'));
99 end
100 singOK = size(solutions, 2) == 8 && singErr <= tol;

```

```

98
99 fprintf('Random test cases:                %d\n', numTests);
100 fprintf('All 8 solutions reproduce target pose:    %d / %d\n', numPoseOK,
    numTests);
101 fprintf('True joint angles found among solutions:    %d / %d\n',
    numRecovered, numTests);
102 fprintf('8 distinct solutions (min dist > %.0e):    %d / %d\n', distTol,
    numDistinct, numTests);
103 fprintf('Smallest pairwise solution distance:        %.3e rad\n',
    minPairDist);
104 fprintf('Worst position error:                    %.3e\n', worstPosErr);
105 fprintf('Worst rotation error (Frobenius norm):        %.3e\n', worstRotErr);
106 fprintf('Singular case (theta5 = 0) pose error:        %.3e\n', singErr);
107
108 fig = figure('Color', 'w');
109 semilogy(1:numTests, max(testPosErr, eps), 'o', 'MarkerFaceColor', [0.1
    0.3 0.7], 'MarkerSize', 4); hold on;
110 semilogy(1:numTests, max(testRotErr, eps), 's', 'MarkerFaceColor', [0.85
    0.4 0.1], 'MarkerSize', 4);
111 yline(tol, '--r', 'Tolerance');
112 ylim([1e-16 1e-6]);
113 xlabel('Test case'); ylabel('Worst error over 8 solutions');
114 legend('Position error (m)', 'Rotation error (Frobenius)', 'Location', '
    best');
115 title('Problem 1: IK verification over 100 random test cases');
116 grid on;
117 exportgraphics(fig, '../Latex/Figures/Problem1ErrorFigure.png', '
    Resolution', 200);
118
119 fid = fopen('../Latex/Problem1Results.tex', 'w');
120 fprintf(fid, '\\begin{tabular}{l r}\n\\hline\n');
121 fprintf(fid, 'Random test cases & %d \\\n', numTests);
122 fprintf(fid, 'All 8 solutions reproduce target pose & %d / %d \\\n',
    numPoseOK, numTests);
123 fprintf(fid, 'True joint angles found among solutions & %d / %d \\\n',
    numRecovered, numTests);
124 fprintf(fid, '8 distinct solutions (min pairwise distance  $>10^{-6}$  rad)
    & %d / %d \\\n', numDistinct, numTests);
125 fprintf(fid, 'Smallest pairwise solution distance (rad) &  $%.2e$  \\\n',
    minPairDist);
126 fprintf(fid, 'Worst position error (m) &  $%.2e$  \\\n', worstPosErr);
127 fprintf(fid, 'Worst rotation error (Frobenius norm) &  $%.2e$  \\\n',
    worstRotErr);
128 fprintf(fid, 'Singular case ( $\\theta_5=0$ ), all solutions reach pose & %s
    \\\n', string(singOK));
129 fprintf(fid, '\\hline\n\\end{tabular}\n');
130 fclose(fid);

```

ikABB.m

Listing 2: ikABB.m

```

1 % ikABB: Closed-form inverse kinematics of the ABB arm.
2 %
3 % q = ikABB(T06, dims)
4 % Returns every joint-angle solution that reaches the pose T06.
5 %
6 % q = [6x8] one solution per column, [theta1; ...; theta6] wrapped to (-pi
  , pi].
7 %   Columns are ordered by (shoulder flip, elbow, wrist). An unreachable
  pose
8 %   returns a 6x0 matrix. At a wrist singularity (sin(theta5) = 0)
  theta4 is set
9 %   to 0, so the two wrist columns of each arm configuration are
  identical.
10 %
11 % T06 = [4x4] desired homogeneous pose of frame 6 in the base frame
12 % dims = [1x5] link parameters [d1, a2, a3, d4, d6]
13 %
14 % Arael Anaya
15 % 10920967
16 % MEGN544
17 % 10/03/2026
18 function q = ikABB(T06, dims)
19     d1 = dims(1); a2 = dims(2); a3 = dims(3); d4 = dims(4); d6 = dims
    (5);
20     clampTol = 1e-9;
21     sinTol = 1e-9;
22
23     R06 = T06(1:3, 1:3);
24     P06 = T06(1:3, 4);
25
26
27     P05 = P06 - d6 * R06(:,3);
28
29     r = sqrt(P05(1)^2 + P05(2)^2);
30     D = sqrt(r^2 + (P05(3)-d1)^2);
31     L = sqrt(a3^2 + d4^2);
32
33     q = zeros(6, 0);
34     if D < eps
35         return;
36     end
37
38     cosBeta = (a2^2 + D^2 - L^2) / (2 * a2 * D);
39     cosGamma = (a2^2 + L^2 - D^2) / (2 * a2 * L);
40
41     if abs(cosBeta) > 1 + clampTol || abs(cosGamma) > 1 + clampTol
42         return;
43     end
44     cosBeta = max(-1, min(1, cosBeta));
45     cosGamma = max(-1, min(1, cosGamma));

```



```

46
47     beta = atan2(sqrt(1 - cosBeta^2), cosBeta);
48     gamma = atan2(sqrt(1 - cosGamma^2), cosGamma);
49     gammaSupp = pi - gamma;
50     delta = atan2(d4, a3);
51
52     q = zeros(6, 8);
53     k = 0;
54     for flip = [0 1]
55         thetaOne = atan2(P05(2), P05(1)) + flip*pi;
56         rSigned = (1 - 2*flip) * r;
57
58         for elbow = [1 -1]
59             thetaTwo = atan2(P05(3)-d1, rSigned) + elbow*beta;
60             thetaThree = -elbow*gammaSupp - delta;
61
62             % OR3 = Rz(th1) Ry(-(th2+th3))
63             phi = thetaTwo + thetaThree;
64             c1 = cos(thetaOne); s1 = sin(thetaOne);
65             R03 = [c1 -s1 0; s1 c1 0; 0 0 1] * ...
66                 [cos(phi) 0 -sin(phi); 0 1 0; sin(phi) 0 cos(phi)];
67             R36 = R03' * R06;
68
69             for wrist = [1 -1]
70                 sinFive = wrist * sqrt(R36(1,3)^2 + R36(2,3)^2);
71                 thetaFive = atan2(sinFive, R36(3,3));
72
73                 if abs(sinFive) < sinTol
74                     % Wrist singularity: only theta4 +/- theta6 is
75                     % determined
76                     thetaFour = 0;
77                     if R36(3,3) > 0
78                         thetaSix = atan2(R36(2,1), R36(1,1));
79                     else
80                         thetaSix = atan2(R36(2,1), -R36(1,1));
81                     end
82                 else
83                     thetaFour = atan2(-R36(2,3) / sinFive, -R36(1,3) /
84                         sinFive);
85                     thetaSix = atan2(-R36(3,2) / sinFive, R36(3,1) /
86                         sinFive);
87                 end
88
89                 k = k + 1;
90                 qk = [thetaOne; thetaTwo; thetaThree; thetaFour; thetaFive
91                     ; thetaSix];
92                 q(:,k) = atan2(sin(qk), cos(qk));
93             end
94         end
95     end
96 end

```

Problem 1 (zero-angle figure)

Listing 3: Problem1Figure.m

```

1  % Problem1Figure: Sketch the zero-angle configuration of the ABB arm from
   % the given DH table.
2  %
3  % Arael Anaya
4  % 10920967
5  % MEGN544
6  % 10/03/2026
7
8  clear; close all; clc;
9
10 addpath(genpath('.../.../Functions'));
11
12 d1 = 0.5;  a2 = 1.0;  a3 = 0.2;  d4 = 1.0;  d6 = 0.3;
13
14 DH = [      0      d1      pi/2      0      ;
15         a2      0      0      0      ;
16         a3      0     -pi/2      0      ;
17         0      d4      pi/2      0      ;
18         0      0     -pi/2      0      ;
19         0      0      0      0      ];
20 DH(6,2) = d6;
21
22 T = fkChain(DH);
23
24 fig = figure('Name', 'Problem 1: Zero-Angle Configuration', 'Color', 'w');
25 hold on; grid on; axis equal;
26 xlabel('X'); ylabel('Y'); zlabel('Z');
27 title('Zero-Angle Configuration of the ABB Arm');
28 view(3);
29
30 origins = squeeze(T(1:3,4,:))';
31 plot3(origins(:,1), origins(:,2), origins(:,3), 'k-o', 'LineWidth', 1.5, '
   MarkerFaceColor', 'k');
32
33 axisLen = 0.25;
34 for i = 1:size(T,3)
35     p = T(1:3,4,i);
36     quiver3(p(1), p(2), p(3), T(1,1,i), T(2,1,i), T(3,1,i), axisLen, 'r',
37             'LineWidth', 1.5);
38     quiver3(p(1), p(2), p(3), T(1,2,i), T(2,2,i), T(3,2,i), axisLen, 'g',
39             'LineWidth', 1.5);
40     quiver3(p(1), p(2), p(3), T(1,3,i), T(2,3,i), T(3,3,i), axisLen, 'b',
41             'LineWidth', 1.5);
42     % Frames 4 and 5 share an origin, so they get one combined label
43     % offset sideways
44     if i == 5
45         text(p(1)+0.04, p(2)+0.2, p(3)+0.04, '\{4\},\{5\}');
46     elseif i ~= 6
47         text(p(1)+0.04, p(2)+0.04, p(3)+0.04, sprintf('\{d\}', i-1));
48     end

```

```
45 end
46
47 exportgraphics(fig, '../Latex/Figures/Problem1Figure.png', 'Resolution',
    200);
```

dhTransform.m

dhTransform.m calls the custom `rotZ` and `rotX` functions below (both take radians), not MATLAB's built-in `rotz/rotx` (which take degrees).

Listing 4: dhTransform.m

```
1 % dhTransform: Homogeneous transform from DH parameters.
2 %
3 % H = dhTransform(a, d, alpha, theta)
4 % Homogeneous transform from DH parameters.
5 %
6 % H = [4x4] homogeneous transform
7 %
8 % a = link length
9 % d = link offset
10 % alpha = link twist (rad)
11 % theta = joint angle (rad)
12 %
13 % Arael Anaya
14 % 10920967
15 % MEGN544
16 % 09/28/2026
17 function H = dhTransform(a, d, alpha, theta)
18     R = rotZ(theta) * rotX(alpha);
19     p = [a*cos(theta); a*sin(theta); d];
20
21     H = [R, p;
22          0, 0, 0, 1];
23 end
```

rotZ.m

Listing 5: rotZ.m

```
1 % rotZ: Rotation matrix about the Z axis.
2 %
3 % R = rotZ(theta)
4 % Rotation matrix about the Z axis.
5 %
6 % R = [3x3] rotation matrix
7 %
8 % theta = rotation angle (rad)
9 %
10 % Arael Anaya
11 % 10920967
12 % MEGN544
13 % 09/19/2026
14 function R = rotZ(theta)
15     c = cos(theta);
16     s = sin(theta);
17     R = [c -s 0;
18         s  c 0;
19         0  0 1];
20 end
```

rotX.m

Listing 6: rotX.m

```
1 % rotX: Rotation matrix about the X axis.
2 %
3 % R = rotX(theta)
4 % Rotation matrix about the X axis.
5 %
6 % R = [3x3] rotation matrix
7 %
8 % theta = rotation angle (rad)
9 %
10 % Arael Anaya
11 % 10920967
12 % MEGN544
13 % 09/19/2026
14 function R = rotX(theta)
15     c = cos(theta);
16     s = sin(theta);
17     R = [1 0 0;
18         0 c -s;
19         0 s  c];
20 end
```

fkChain.m

Listing 7: fkChain.m

```

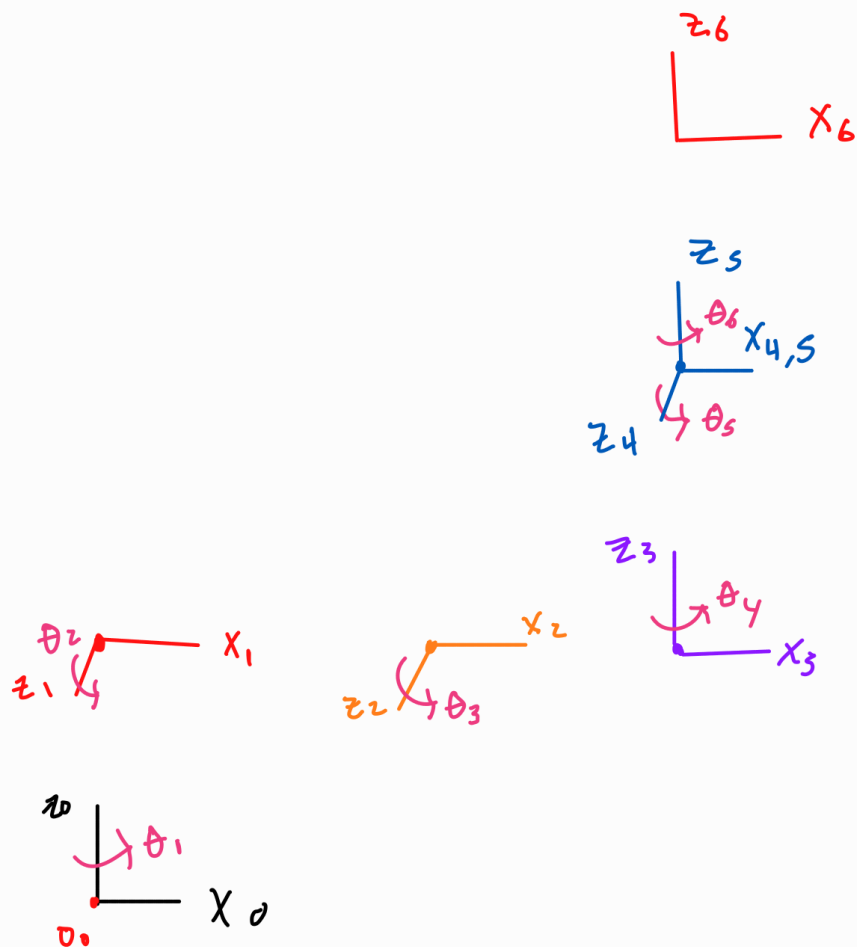
1 % fkChain: Cumulative forward-kinematics transforms from a DH table.
2 %
3 % T = fkChain(DH)
4 % Cumulative forward-kinematics transforms from a DH table.
5 %
6 % T = [4x4x(n+1)] stack of transforms, T(:,:,1) = base frame,
7 %     T(:,:,i+1) = 0Ti for link i
8 %
9 % DH = [n x 4] table of [a, d, alpha, theta] rows
10 %
11 % Arael Anaya
12 % 10920967
13 % MEGN544
14 % 09/28/2026
15 function T = fkChain(DH)
16     n = size(DH, 1);
17     T = zeros(4, 4, n+1);
18     T(:,:,1) = eye(4);
19     for i = 1:n
20         Ti = dhTransform(DH(i,1), DH(i,2), DH(i,3), DH(i,4));
21         T(:,:,i+1) = T(:,:,i) * Ti;
22     end
23 end

```

Appendix B: Hand Calculations

Link	a	d	α	θ
1	0	d_1	$\pi/2$	θ_1
2	a_2	0	0	θ_2
3	a_3	0	$-\pi/2$	θ_3
4	0	d_4	$\pi/2$	θ_4
5	0	0	$-\pi/2$	θ_5
6	0	d_6	0	θ_6

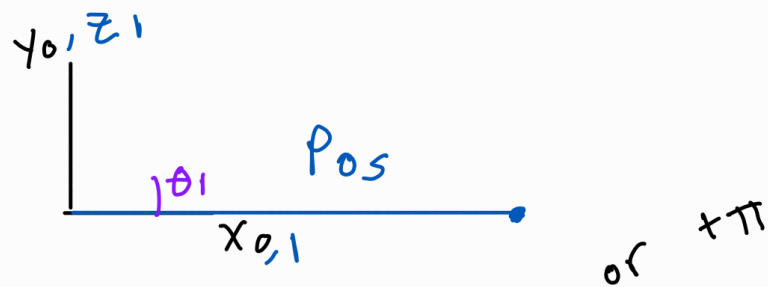
a)



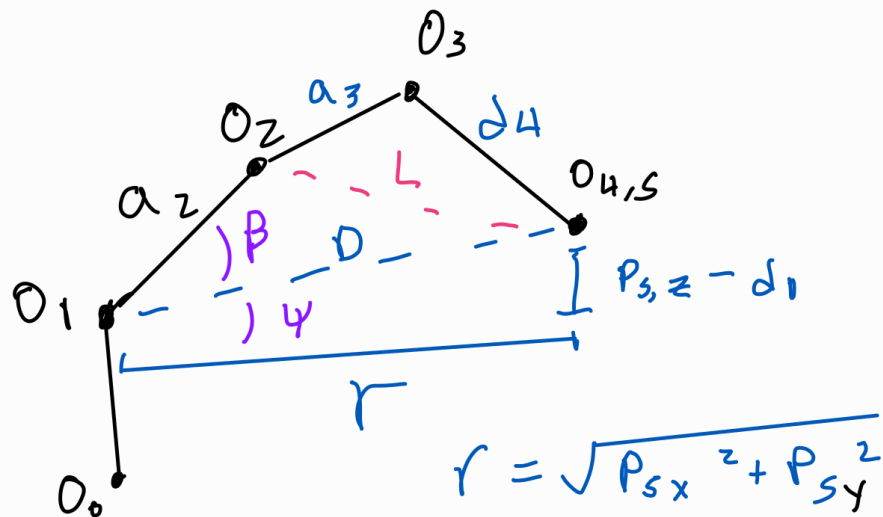
b)

$$\text{Let } P_{0,6} = [P_{6x}, P_{6y}, P_{6z}]^T$$

$$P_{05} = P_{06} - d_6 \cdot R(:, 3)$$



$$\therefore \theta_1 = \text{atan}(P_{05,y}, P_{05,x})$$



$$\psi = \text{atan2}(P_{s,z} - d_1, \sqrt{P_{sx}^2 + P_{sy}^2})$$

$$C^2 \dot{\theta}^2 = 1$$

$$\theta_2 = \psi \pm \beta$$

$$L^2 = a_2^2 + D^2 - 2a_2 D C(\beta)$$

$$D = \sqrt{\sqrt{P_{sx}^2 + P_{sy}^2} + (P_{sz} - d_1)^2}$$

$$L = \sqrt{a_3^2 + d_4^2}$$

$\therefore$

$$C(\beta) = \frac{a_2^2 + D^2 - L^2}{2a_2 D}$$

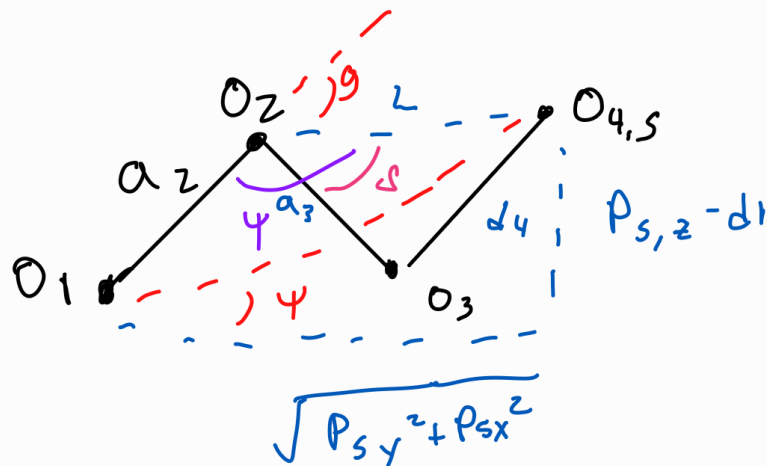
$$S(\beta) = \pm \sqrt{1 - C^2(\beta)}$$

$$\beta = \arctan 2 \left(\pm \sqrt{1 - \frac{L^2 - a_2^2 - D^2}{2a_2 D}}, \frac{L^2 - a_2^2 - D^2}{2a_2 D} \right)$$

$$\theta_z =$$

$$\arctan 2 \left(p_{s,z} - d_1, \sqrt{p_{sx}^2 + p_{sy}^2} \right)$$

$$+ \arctan 2 \left(\sqrt{1 - \left(\frac{a_z^2 + D^2 - L^2}{2a_z D} \right)^2}, \frac{a_z^2 + D^2 - L^2}{2a_z D} \right)$$



$$S = \arctan 2 (d_4, a_3)$$

$$\theta = \pi - \gamma$$

$$\theta_3 = \theta - \delta = (\pi - \gamma) - \delta$$

$$D^2 = a_2^2 + L^2 - 2a_2L \cos(\gamma)$$

$$\therefore \cos \gamma = \frac{a_2^2 + L^2 - D^2}{2a_2L}$$

$$\sin \gamma = \sqrt{1 - \cos^2 \gamma}$$

$$\gamma = \arctan \left(\frac{\sqrt{1 - \cos^2 \gamma}}{\cos \gamma} \right)$$

$\therefore$

$$\theta_3 =$$

$$\pi - \arctan \left(\frac{\sqrt{1 - \frac{a_2^2 + L^2 - D^2}{2a_2L}}}{\frac{a_2 + L^2 - D^2}{2a_2L}} \right)$$

$$- \arctan \left(\frac{D}{a_2} \right)$$

Find 0R_3 from DH table

$${}^3R_6 = {}^0R_3^{-1} \cdot {}^0R_6$$

This is $z y z$

$\therefore$

$$\theta_5 = \text{atan2} \left(\pm \sqrt{r_{13}^2 + r_{23}^2}, r_{33} \right)$$

$$\theta_4 = \text{atan2} \left(\frac{-r_{23}}{s\theta_5}, \frac{-r_{13}}{s\theta_5} \right)$$

$$\theta_6 = \text{atan2} \left(\frac{-r_{32}}{s\theta_5}, \frac{r_{31}}{s\theta_5} \right)$$